RV Institute of Technology and Management®

Department of Master of Computer Applications

LITERATURE SURVEY

AI Emergency Response Platform

1. Introduction

Reporting an emergency or incident is often slow and error-prone when it depends on phone calls, text messages or manual records. Reporters may not know or be able to describe their exact location, descriptions are inconsistent, and the person who reported has little visibility into what happens next. Several researchers have built mobile applications that send an incident type, description, photo and GPS location to responders, while separate studies have applied natural language processing (NLP) and machine learning to classify emergency incident descriptions.

This survey studies that work in relation to the proposed **AI Emergency Response Platform**: a web application where a user reports an incident with a description, an optional image and a location (GPS, map or manual text); an AI component *recommends* an incident category and priority; the report is routed to an authorised responder; and the user can track the status until resolution. The AI output is advisory only and is verified by a human responder or administrator. The platform is a college demonstration project and does not contact real emergency services.

2. Scope and Method of the Survey

The survey covers five areas relevant to the project: (a) mobile and web incident reporting, (b) GPS and map-based location reporting, (c) crowdsourced disaster reporting, (d) AI/NLP-based incident classification and triage, and (e) AI-assisted emergency management with human oversight. Each paper was checked against its publisher, repository or DOI page. Only details that appear in the paper or its official abstract page are reported here. The reading depth for every paper is stated (full text or abstract only), and the preprint (Paper 7) is labelled as not yet peer-reviewed. This is a starting survey of seven papers; it will be expanded as the project proceeds.

3. Summary of Reviewed Papers

Paper 1. Mobile Application for Incident Reporting	
Authors and year	Mary Ann E. Ignaco (Cavite State University, Philippines), 2021
Publication	JOIV: International Journal on Informatics Visualization, vol. 5, no. 4, pp. 388–394
Problem addressed	In the Philippines, incident reporting depends on visiting the nearest law-enforcement office or calling the 911 hotline, which cannot immediately retrieve the reporter's location unless the reporter describes it clearly.
Method / technology	Android app (Java, Android Studio, Firebase, Google Maps) with a web admin site, built using the Agile model. The app captures GPS latitude/longitude automatically or lets the user pin a location on the map, plus incident type (public disturbance, ordinance violation, crime), description and photos. The report goes to the nearest barangay responder, who can request backup (ambulance, fire station) or pass it to the police. Report status: Pending, Responded, On-going, Closed.
Main findings	Evaluated by 30 respondents using the ISO 25010 model: overall mean 4.61 ("Excellent"); functionality 4.47 ("Very Good"). All 26 listed functional test cases passed. The web admin generates statistical reports and charts.
Limitations / gaps	The paper has no separate limitations section. It describes three fixed incident types and an Android-only client; no AI-based analysis of report text or images is described (observation).
Relevance to this project	Closest in workflow to this project (GPS or manual map pin, photo, description, responder, status tracking, admin statistics). This project adds a web interface and AI-recommended category and priority.
Verified link / access	DOI: 10.30630/joiv.5.4.741. Full-text PDF: free on joiv.org (https://joiv.org/index.php/joiv/article/view/741).
Reading depth	Full text read.

Paper 2. Operationalizing crowdsourcing through mobile applications for disaster management in India	
Authors and year	Vibhas Sukhwani and Rajib Shaw, 2020
Publication	Progress in Disaster Science, vol. 5, Article 100052
Problem addressed	Lack of real-time data on emerging disasters limits decision makers, especially in developing countries such as India. App-based crowdsourcing of disaster information is proposed as a remedy.
Method / technology	Systematic investigation of the current state of 33 freely accessible disaster-related mobile apps in India.
Main findings	Most of the 33 apps are primarily educational and their outreach is highly limited. The authors suggest enhancing community outreach, ensuring a user-friendly interface, and promoting GPS-based apps.
Limitations / gaps	Authors report limited outreach and the dominance of educational apps. The study surveys apps; it does not evaluate AI classification or responder workflows (observation).
Relevance to this project	Supports the design choices of a simple, user-friendly reporting form with GPS support, and motivates a platform focused on reporting and response rather than information only.
Verified link / access	DOI: 10.1016/j.pdisas.2019.100052. Full text: open-access journal; author-hosted PDF at https://rajibshaw.org/wpRS/wp-content/uploads/2019/11/Sukhwani-Shaw-PDSIS-.pdf
Reading depth	Official abstract and published summaries read; full text available to read.

Paper 3. Improving the usability of large emergency 911 data reporting systems: A machine learning case study using emergency incident descriptions	
Authors and year	N. Katherine Yoon, Tyler D. Quinn, Alexa Furek, Nora Y. Payne, Emily J. Haas, 2025
Publication	Journal of Safety Research, vol. 93, pp. 335–341
Problem addressed	Emergency 9-1-1 incident data are recorded voluntarily in department-specific dispatch systems. Inconsistent reporting practices reduce data quality, and a broad three-category scheme (EMS, Fire, Other) forced some incidents into wrong categories.
Method / technology	3,564 unique incident descriptions from six fire departments were manually recoded into nine categories. Text was preprocessed (stop-word removal, lowercasing, lemmatization) and converted with TF-IDF and n-grams. Naive Bayes, Random Forest and SVM were trained (75% train / 25% test).
Main findings	Random Forest with unigrams reached 93% accuracy (F1 = 0.90), SVM 92% (F1 = 0.89), Naive Bayes 85% (F1 = 0.61). Most errors involved incidents predicted as EMS or Other/Unclear.
Limitations / gaps	Authors note possible bias from manual coding, many near-duplicate descriptions, EMS dominating the data (58.2%), and department-specific terms that may not generalise; manual coding is recommended for new departments.
Relevance to this project	Shows that text-based incident classification is feasible with simple, explainable models and gives the evaluation metrics to reuse (accuracy, precision, recall, F1, confusion matrix). It studies dispatch descriptions, not a citizen reporting and routing platform.
Verified link / access	DOI: 10.1016/j.jsr.2025.04.001. Full-text PDF: free via CDC Stacks (https://stacks.cdc.gov/view/cdc/208965).
Reading depth	Full text read.

Paper 4. Design and Development of an Incident Reporting Mobile Application	
Authors and year	Jan Peter Bulusan, Bradd Melton Cabaluna, Daniel John Macaraig, Abner Cruz (Adventist University of the Philippines), 2019
Publication	11th International Scholars Conference (Abstract Proceedings), vol. 7, no. 1, pp. 1751–1762
Problem addressed	Reporting incidents through text messages and phone calls is tedious and prone to errors and misinterpretation.
Method / technology	Mobile application that transmits incident type, location and images, developed with the prototyping model. Target users were 38 trained first-responder motorcycle riders in Silang, Cavite; interviews and benchmarking of related systems were used.
Main findings	Users described the app as accurate, less error-prone and easy to use. GPS location can be accurate to within about five metres, the receiver can view the location on a map, and reports are stored centrally.
Limitations / gaps	Small and specific user group (38 riders). The authors suggest future data mining of stored reports for hotspots and peak times, and using AI to identify and request help from the nearest medical facilities for larger areas.
Relevance to this project	Supports minimal text entry, GPS plus map view, and image attachment. Its AI suggestion matches this project’s direction, but this project keeps a human responder in control instead of automatically requesting help.
Verified link / access	DOI: 10.35974/isc.v7i1.1573. Full-text PDF link listed on the conference site (https://jurnal.unai.edu/isc/article/view/1573).
Reading depth	Abstract and publication page read; full text not opened.

Paper 5. Design, Implementation, and Evaluation of a Mobile Application for Real-Time Incident Management in Local Government	
Authors and year	Marleny Peralta Ascue, Luis Bryan Caballero Pinto, José Luis Poluco Sicos, José Luis Merma Aroni, 2026
Publication	International Journal of Advanced Computer Science and Applications (IJACSA), vol. 17, no. 7, pp. 285–292
Problem addressed	Without an integrated digital tool, local governments in Peru find it hard to identify the location, time and frequency of incidents in real time; registration relied on manual records and phone calls.
Method / technology	Flutter mobile app, PHP REST API and MySQL, built with Agile Scrum. The user picks a category (public safety, environmental, traffic, civil protection), adds a description, photo/video and GPS location; the system validates the form, assigns the case to the matching municipal department, sends notifications, and tracks status (pending, in progress, resolved). Evaluated with a pretest–posttest design and the CSUQ usability questionnaire.
Main findings	Average registration time fell by 99.26% (154.23 to 1.15 minutes) and response time by 31.68% (94.69 to 64.69 minutes); overall CSUQ satisfaction was 6.15 out of 7 (Cronbach's alpha 0.948).
Limitations / gaps	Single municipality. The authors list AI techniques for incident prioritisation and predictive analysis as future work.
Relevance to this project	Closest match to the workflow of this project (report, category, assignment, status, dashboard) and a good model for evaluation. It confirms that AI-based prioritisation is still described as future work.
Verified link / access	Full-text PDF: free on the publisher site (https://thesai.org/Downloads/Volume17No7/Paper_27-Design_Implementation_and_Evaluation_of_a_Mobile_Application.pdf).
Reading depth	Full text read.

Paper 6. Human-AI Use Patterns for Decision-Making in Disaster Scenarios: A Systematic Review	
Authors and year	Emmanuel Adjei Domfeh and Christopher L. Dancy, 2025
Publication	2025 IEEE International Symposium on Technology and Society (ISTAS 2025); arXiv:2509.12034
Problem addressed	Disaster decisions are made under uncertainty with limited resources; the paper asks how human–AI collaboration can support decision-making across disaster phases.
Method / technology	Systematic review of 51 peer-reviewed studies, grouped into Human-AI decision support systems, task and resource coordination, trust and transparency, and simulation and training.
Main findings	AI can improve situational awareness, response efficiency and support for complex decisions, but limitations remain in scalability, interpretability and system interoperability.
Limitations / gaps	Authors call for adaptive, trustworthy and context-aware human–AI systems.
Relevance to this project	Justifies the design rule that AI only recommends and an authorised human decides, and the use of AI summary and reason fields so recommendations can be explained.
Verified link / access	arXiv: https://arxiv.org/abs/2509.12034 (free).
Reading depth	Abstract read.

Paper 7. Automated Triaging and Transfer Learning of Incident Learning Safety Reports Using Large Language Representational Models	
Authors and year	Peter Beidler and five co-authors (full list on the arXiv page), 2025
Publication	arXiv preprint 2509.13706 (cs.CL), not yet peer-reviewed
Problem addressed	Large repositories of free-text safety incident reports need automatic screening to find high-severity reports (studied in radiation oncology across two institutions).
Method / technology	Baseline support vector machines compared with BlueBERT, a language model pretrained on PubMed abstracts and hospital patient data; a transfer-learning variant was tested on another institution's data.
Main findings	The authors report successfully developing cross-institution NLP models for incident report text.
Limitations / gaps	Preprint; medical-safety reports rather than emergency or citizen reports.
Relevance to this project	Shows that severity (priority) screening from free text is an established NLP task and that models can transfer across institutions, supporting priority recommendation with human review.
Verified link / access	arXiv: https://arxiv.org/abs/2509.13706 (free).
Reading depth	Abstract read.

4. Literature Survey Table

S. No.	Author and Year	Paper Title	Problem Addressed	Method or Technology	Main Findings	Limitations / Research Gap	Relevance to Our Project
1	Ignaco, 2021	Mobile Application for Incident Reporting	Location of reporters not available quickly to responders in 911 / police reporting	Android app, Firebase, Google Maps; GPS or manual pin, photo, type; Agile	ISO 25010 mean 4.61 (30 respondents); 26 test cases passed	Fixed incident types; Android only; no AI analysis described	Closest workflow: GPS/map, photo, responder, status, admin stats
2	Sukhwani and Shaw, 2020	Operationalizing crowdsourcing through mobile applications for disaster management in India	Lack of real-time disaster data; potential of app-based crowdsourcing	Review of 33 free disaster apps in India	Mostly educational apps; outreach highly limited; recommends user-friendly, GPS-based apps	Limited outreach; surveys apps only, no AI or responder workflow	Supports simple GPS-enabled reporting interface
3	Yoon et al., 2025	Improving the usability of large emergency 911 data reporting systems: A machine learning case study	Inconsistent 9-1-1 incident categories and data quality	TF-IDF, n-grams; Naive Bayes, Random Forest, SVM on 3,564 descriptions	Random Forest 93% accuracy (F1 0.90); errors mostly toward EMS	Manual-coding bias; EMS-dominated data; regional terms	Feasibility and metrics for text-based classification
4	Bulusan et al., 2019	Design and Development of an Incident Reporting Mobile Application	SMS / phone incident reporting is tedious and error-prone	Mobile app with type, location, image; prototyping; 38 first responders	Accurate, easy to use; GPS within about 5 m; map view for receiver	Small user group; AI only suggested for future	Minimal typing, GPS plus map, image upload
5	Peralta Ascue et al., 2026	Design, Implementation, and Evaluation of a Mobile Application for Real-Time Incident Management in Local Government	No integrated tool to locate and manage citizen incidents in real time	Flutter, PHP REST API, MySQL; Scrum; pretest-posttest, CSUQ	Registration time -99.26%; response time -31.68%; CSUQ 6.15/7	Single municipality; AI prioritisation left as future work	Closest end-to-end workflow; evaluation model
6	Domfeh and Dancy, 2025	Human-AI Use Patterns for Decision-Making in Disaster Scenarios: A Systematic Review	How human-AI collaboration supports disaster decisions	Systematic review of 51 studies	AI helps awareness and efficiency; limits in interpretability and scalability	Need for trustworthy, context-aware human-AI systems	AI as recommendation; human final decision; explanations
7	Beidler et al., 2025 (preprint)	Automated Triaging and Transfer Learning of Incident Learning Safety Reports Using Large Language Representational Models	Screening high-severity incident reports from free text	SVM vs BlueBERT; transfer learning across two institutions	Cross-institution NLP models developed successfully (authors)	Preprint; radiation-oncology reports, not emergencies	Severity / priority triage from text is established

5. Existing System Comparison

5.1 Features documented in the reviewed systems

“Not described” means the feature is not mentioned in the paper or abstract that was read; it does not prove the system lacks it. Paper 3 is a classification study, not a reporting application, so it is compared only on AI analysis.

Feature	Ignaco (2021)	Bulusan et al. (2019) abstract only	Peralta Ascue et al. (2026)	Yoon et al. (2025)	Proposed platform
Text description	Yes	Not stated in abstract	Yes	Input data (dispatch text)	Yes
Photo / image upload	Yes	Yes	Yes (photo and video)	N/A	Yes, optional
GPS location	Yes, automatic	Yes (about 5 m)	Yes	N/A	Yes
Manual map pin / typed location	Yes, pin on map	Not stated in abstract	Not described	N/A	Yes: map click and typed description
Incident category	Reporter chooses from 3 types	Incident type sent by reporter	Reporter chooses from 4 categories	ML assigns 9 categories (offline study)	AI recommends; human verifies
AI / ML analysis of the report	Not described	Suggested as future work	Not described (future work)	Yes, Naive Bayes / RF / SVM	Yes, planned (model to be selected)
Priority recommendation	Not described	Not described	Not described (AI prioritisation is future work)	Not described	Yes: Low / Medium / High / Critical
Responder assignment	Yes, nearest barangay station	Report receiver; details not stated	Yes, by category to department	N/A	Routing by category plus admin reassignment
Status tracking for the reporter	Yes (4 statuses)	Not stated in abstract	Yes (3 statuses)	N/A	Yes: timeline and history
Admin analytics	Yes, charts and reports	Data mining suggested as future work	Yes, statistics dashboard	N/A	Yes, with filters
Client type	Android app	Mobile app	Flutter mobile app	N/A	Responsive web app

The “Proposed platform” column describes planned features of a project that is still in the design stage; none are claimed as implemented.

5.2 What existing systems provide or lack, and what the proposed platform adds

What the reviewed systems provide or do not document	What the proposed AI Emergency Response Platform intends to add
Citizen reporting with type, description, photo and GPS is already documented in several mobile apps (Papers 1, 4, 5).	The same core reporting, offered in one responsive web application with three location options: current GPS, map click, or a typed description.
Incident category is chosen by the reporter or limited to fixed types (Papers 1, 5).	AI suggests a category (Fire, Medical, Security, Accident, Infrastructure, Maintenance, Other) from the description and optional image, so the reporter does not need to know the category.
Priority recommendation is not described in the three reporting apps; Paper 5 lists AI prioritisation as future work.	An AI-recommended priority (Low, Medium, High, Critical) with a short summary and reason, shown as a recommendation.
ML text classification has been shown to work on dispatch descriptions (Paper 3: 93% accuracy with Random Forest) but as a data-standardisation study, separate from a reporting and routing platform.	Classification inside the live reporting workflow, with backend validation of the AI output and a fallback (report stays saved and is reviewed manually) if the AI service fails.
Routing is by nearest station (Paper 1) or by chosen category (Paper 5).	Routing by AI-recommended category to the matching responder group, with administrator reassignment and a “Pending Assignment” state when no responder is available.
Status tracking and dashboards exist in Papers 1 and 5.	Status timeline with an audit history of every change, in-app notifications, and role-based dashboards with analytics filters.
Paper 6 stresses trust, transparency and interpretability for human–AI systems in disasters.	AI is advisory only: results are labelled as recommendations, administrators can correct them, and the system never contacts real emergency services.
Paper 2 found that many disaster apps are educational with limited outreach.	A short, fast report form that works on mobile browsers and avoids unnecessary fields.

6. Research Gap

Within the seven sources reviewed, the following points are supported:

- Incident reporting applications with GPS or map location, photos, responder assignment, status tracking and admin statistics already exist and have been evaluated (Papers 1, 4, 5).
- In these applications the incident type is selected by the reporter or fixed in advance, and none of the three describes AI-based category or priority analysis. Two of them name AI as future work (Paper 4 for requesting help from nearby facilities, Paper 5 for prioritisation).
- Text-based machine-learning classification of emergency incident descriptions has been demonstrated with strong accuracy (Paper 3) and severity screening from free text is an established task (Paper 7), but these studies are carried out on existing records, separately from a citizen reporting, routing and tracking platform.
- Research on human–AI collaboration in disasters stresses trust, transparency and human oversight (Paper 6).

Gap statement. The reviewed work does not describe a single system that combines citizen incident reporting with multiple location options, AI-assisted category and priority recommendation, human-verified routing to responders, and reporter-visible status tracking. The proposed AI Emergency Response Platform aims to integrate these capabilities in one web application. This is an integration contribution only: other commercial or government systems that were not reviewed here may already provide some of these features, and this survey will be extended to check that.

7. Conclusion

The literature shows that GPS-based incident reporting with photos, responder handling and status tracking is practical and well received by users, and that machine learning can classify emergency incident text with high accuracy. It also shows that AI-based prioritisation is still treated as future work in reporting applications and that human oversight is considered essential when AI supports emergency decisions. These findings guide the design of the AI Emergency Response Platform: a simple multi-option reporting form, an explainable and validated AI recommendation of category and priority, routing to authorised responders, and a transparent status timeline, with the final decision always left to a human.

8. References

[1] M. A. E. Ignaco, "Mobile Application for Incident Reporting," *JOIV: International Journal on Informatics Visualization*, vol. 5, no. 4, pp. 388–394, Dec. 2021, doi: 10.30630/joiv.5.4.741.

[2] V. Sukhwani and R. Shaw, "Operationalizing crowdsourcing through mobile applications for disaster management in India," *Progress in Disaster Science*, vol. 5, Art. no. 100052, 2020, doi: 10.1016/j.pdisas.2019.100052.

[3] N. K. Yoon, T. D. Quinn, A. Furek, N. Y. Payne, and E. J. Haas, "Improving the usability of large emergency 911 data reporting systems: A machine learning case study using emergency incident descriptions," *Journal of Safety Research*, vol. 93, pp. 335–341, 2025, doi: 10.1016/j.jsr.2025.04.001.

[4] J. P. Bulusan, B. M. Cabaluna, D. J. Macaraig, and A. Cruz, "Design and Development of an Incident Reporting Mobile Application," *11th International Scholars Conference*, vol. 7, no. 1, pp. 1751–1762, 2019, doi: 10.35974/isc.v7i1.1573.

[5] M. Peralta Ascue, L. B. Caballero Pinto, J. L. Poluco Sicos, and J. L. Merma Aroni, "Design, Implementation, and Evaluation of a Mobile Application for Real-Time Incident Management in Local Government," *International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 17, no. 7, pp. 285–292, 2026.

[6] E. A. Domfeh and C. L. Dancy, "Human-AI Use Patterns for Decision-Making in Disaster Scenarios: A Systematic Review," in *Proc. 2025 IEEE International Symposium on Technology and Society (ISTAS)*, 2025; arXiv:2509.12034.

[7] P. Beidler *et al.*, "Automated Triaging and Transfer Learning of Incident Learning Safety Reports Using Large Language Representational Models," arXiv:2509.13706, 2025 (preprint, not peer-reviewed).

Appendix: Papers Suggested for Printing

No.	Paper	Free full text	Where
1	Ignaco (2021)	Yes	joiv.org article page
2	Sukhwani and Shaw (2020)	Yes	Author-hosted PDF (rajibshaw.org)
3	Yoon et al. (2025)	Yes	CDC Stacks
5	Peralta Ascue et al. (2026)	Yes	thesai.org (IJACSA)
6	Domfeh and Dancy (2025)	Yes	arXiv 2509.12034
4	Bulusan et al. (2019)	PDF link listed	Conference site (jurnal.unai.edu)